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Exploring veterinary knowledge extension methods and perceived barriers among Canadian dairy veterinarians and producers: A qualitative focus group study

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ABSTRACT

This qualitative study explored veterinarians' and producers' perspectives on the role veterinarians play in knowledge transfer and translation (KTT) for dairy producers and the barriers to success of current KTT strategies. Five focus groups were conducted with 20 dairy producers in Ontario, Canada, and 5 focus groups were conducted with 20 bovine veterinarians across Canada. Using applied thematic analysis, 4 main themes were identified: (1) educational role of the veterinarian, (2) progress is a collaborative approach toward success, (3) veterinary strategies for knowledge transfer and translation, and (4) veterinary barriers to knowledge transfer and translation. Participants felt that the veterinary role is shifting from clinical practitioner to farm advisor, with veterinarians valued for the novel information they can provide. Additionally, it was emphasized that veterinarians serve as an unbiased information filter to ensure their clients are able to make practical and informed management decisions. However, the disconnection of communication between producers and veterinarians was reported by both groups to be a barrier to KTT. Earning client respect was vital to the success of the producer-veterinarian relationship, resulting in trust and ultimately leading to effective and engaged knowledge extension. Both sets of participants discussed that producer success was a collaborative effort that required the adoption of progressive best management practices, for which effective KTT is crucial. Both small group and large annual meetings hosted by veterinarians or in collaboration with other on-farm advisors were considered valuable, but identifying ways to individualize KTT will further engage producers and may alleviate advisor fatigue, which was a barrier reported by veterinarians. Additional KTT barriers include a need to strengthen veterinary emotional intelligence and inconsistencies of advice and practices between veterinarians, both of which result in reduced producer trust and reduced uptake of recommendations. Therefore, identifying and implementing solutions to KTT barriers for both producers and veterinarians will optimize the successful and timely adoption of best management practices.

Key words: knowledge mobilization, communication, knowledge transfer

INTRODUCTION

Over the past decades, the Canadian dairy sector has evolved rapidly (Barkema et al., 2015; von Keyserlingk et al., 2024), and the industry continues to change in response to technological advances and economic pressures (Jelinski et al., 2015, 2020). While dairy producers have many opportunities to learn and to obtain information, it has been reported that it can be difficult to practically interpret and implement findings (Carter et al., 2026). As such, identifying and enhancing the strategies used to convey new results to end users in the industry is essential.

Veterinarians are one of the most influential and trusted sources of information for dairy producers when it comes to animal health (Gerber et al., 2020; Cobo-Angel et al., 2021). Given their influence, they are well-positioned to support the delivery and uptake of educational programs. Despite this, opportunities for veterinarians to serve as educators are often not fully realized (Cobo-Angel et al.,

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

2021; Farrell et al., 2023). The specific barriers, whether physical or social, to knowledge transfer and translation (**KTT**) between veterinarians and dairy producers remain unclear. Dairy producers have reported that hands-on and face-to-face learning is preferred, particularly in a small group and interactive setting (Carter et al., 2026). Although research has examined what dairy producers favor (Carter et al., 2026), less is known about how veterinarians prefer to communicate information, as well as how they currently adapt their education strategies to meet their clients' needs. Gaining insight into perspectives from producers and veterinarians is essential to improve education strategies for effective communication and knowledge exchange between veterinarians and dairy producers.

The objective of this qualitative study was to gain insight into producers' and veterinarians' perspectives on KTT strategies used by veterinarians and the barriers to effective knowledge transfer and implementation. The findings from this study will support the optimization of KTT strategies for veterinarians to suit both their needs and their clients'.

MATERIALS AND METHODS

Research Approach

The collaborative approach fostered through qualitative research methods can provide a rich understanding of the perceived barriers of KTT from veterinarians to dairy producers (Raedts et al., 2017; Belage et al., 2019; Roche et al., 2019). Focus groups were chosen to gain an in-depth understanding of perceptions, opinions, and experiences of dairy producers and veterinarians by stimulating discussion among participants (Krueger and Casey, 2009; Nyumba et al., 2018). A critical realist paradigm was used for this study, which assumes that the uptake of educational programs is influenced by both individual internal perceptions and underlying external factors (Fletcher, 2017). This study was approved by the research ethics board of the University of Guelph (REB #24-03-020 and REB #0142). The participants were provided with information on the study's objectives and methods, as well as any potential individual or industry-wide impacts associated with the research. Informed written consent was obtained from each participant before each session commenced. Participants were also informed of their right not to answer any questions that made them feel uncomfortable or to leave the session at any time; none of the participants did so. The Standards for Reporting Qualitative Research guidelines were used to report the findings (O'Brien et al., 2014).

Positionality and Reflexivity Statement

All members of the research team possess professional knowledge in dairy health, and HC (she/her), CR (she/her), EM (she/her), JDR (she/her), JPR (he/him), SL (he/him), and DR (he/him) hold advanced degrees in epidemiology. Five team members (DR [PhD], CR [PhD], JPR [PhD], SL [DVSc], and JDR [PhD]) are veterinarians. Authors HC (MSc) and EM (MSc) are both PhD candidates, and all members hold prior experience leading qualitative research projects. In this study, the first author (HC) moderated all focus groups but one, which was moderated by JDR in French with francophone participants. Authors HC and EM have experience working closely with Canadian dairy producers and have conducted and disseminated research through their respective master's programs. Through their experiences, they were familiar with common barriers to uptake of educational programs from the perspectives of dairy producers but less so the perspectives of veterinarians. Assumptions and perspectives identified during the analysis and preparation of the manuscript were considered. Through the research process, an affinity for KTT was acknowledged. A research log was maintained during the data analysis to document decision-making rationales and interpretation processes.

Participants

Ten focus groups (40 participants) in total were completed between July 2024 and April 2025. Five face-to-face focus groups were conducted in Ontario, Canada, each with 3 to 5 dairy producers (20 participants) in private and quiet community hall boardrooms to minimize the risks related to confidentiality and to ensure audio recordings were clear. The remaining 5 focus groups were conducted virtually with groups of 3 to 5 bovine or mixed-practice veterinarians across Canada (20 participants). While producer recruitment was accomplished provincially within Ontario, Canada, veterinarian recruitment was conducted nationally to achieve an adequate number of participants. Producers with at least 25 lactating dairy cows and veterinarians with experience providing services to dairy herds were eligible to be included in the study. Producer recruitment was conducted using purposive sampling by having veterinary practitioners provide researcher emails to clients whom they believed would be interested and with snowball sampling, where participants were asked to invite other dairy producers who may be interested in participating. An invitation to participate was also posted on the Dairy at Guelph website; however, this method did not recruit any participants. Bovine veterinarians were recruited to participate through voluntary response and purposive sampling.

Email invitations to participate were sent via listservs for the Canadian Association of Bovine Veterinarians and the Ontario Association of Bovine Practitioners. Approximately 750 veterinarians received invitations through a listserv email, and 18 responded and participated. Industry connections of the research team were also personally invited to participate; 5 were invited, and 2 participants were recruited in this fashion. Although participating veterinarians were encouraged to share the invitation to participate with peers, no participants were recruited using this method. All participants were aware of the researcher's previous work, and some had previous industry interaction with HC and the principal investigator, DR. No participants opted to drop out of the study.

Focus Group Discussion

Before data collection, 2 semistructured discussion guides were created and iteratively reviewed by the research team (Tables 1 and 2) for the producer and veterinarian focus groups. The discussion guide for the focus group with French-speaking veterinarians was translated by the bilingual moderator (JDR), who met with HC to validate the translation. The guide was used as a framework to ensure the research question was fully addressed. Probing questions were used as needed to obtain detailed responses to the topic being discussed. Following the first veterinary focus group, one question that did not elicit well-rounded responses was amended with an added follow-up question to reduce the need for probing. No amendments were made to the producer discussion guide following the first producer focus group. The questions were generated to encourage discussion related to the perception of the role veterinarians play in KTT and strategies and barriers to KTT between veterinarians and dairy producers. The discussion guide for producers was designed to allow for broad and open-ended discussion; therefore, KTT from veterinarians was not specifically targeted through individual questions. Their role in knowledge mobilization emerged organically through participant discussion across multiple questions without probing. Before the focus groups started, each participant completed a short written survey. Producers were asked to share their age, gender, education level, herd size, and production system, and veterinarians were asked to provide information on their age, gender, time practicing, type of practice, and province. A single moderator facilitated each focus group. Author HC moderated 9 focus groups in English, and JDR moderated one focus group in French, with sessions lasting from 40 to 90 min, with an average of 67 min. Notes were taken in all 10 focus groups to record any nonverbal cues. Each session was audio recorded and transcribed using Otter.Ai (https://otter.ai, Mountain View, CA, accessed May 15, 2025) and checked for accuracy of speech and identification by HC through comparison to the original audio files. The one French transcript was translated using ChatGPT (Version GPT-5, Open AI, San Francisco, CA, accessed May 15, 2025) and was also verified for accuracy along with the other 9 transcripts through a comparison to the original audio recordings by a bilingual research team member (HC). Each participant was assigned a participant code to replace their name in the transcript to maintain anonymity, where the first number was randomly assigned as their number within their session, and the second number identified the session in which they participated. The letter "P" indicates producers, and "V" veterinarians. Square brackets were used within quotes for added clarification, and ellipses indicate omission of irrelevant sections within the quotation.

Data Analysis

The data analysis was led by HC, with codebook validation completed by EM. Patterns in the data were identified, analyzed, and reported through applied thematic analysis (Guest et al., 2012). Familiarization with the data was completed through multiple readings of the transcripts and listening to the audio recordings. With an inductive approach, the first author (HC) and a second researcher (EM) coded the 10 transcripts to develop potential themes and subthemes from the data using NVivo (version 14, Lumivero) and Quirkos (version 2.5.3) software. Both HC and EM coded the first 2 producer and veterinarian focus group transcripts to develop a coding scheme (Guest et al., 2012). Their codebooks were then compared in a meeting between HC and EM, where discrepancies were discussed and the codebook was revised. This version of the codebook with labels, descriptions, and sample quotations from the data was used by HC and EM, who both coded the remaining 6 transcripts. After coding was completed, HC and EM met 2 more times to agree on the final codebook. The results were then discussed with DR and CR. Any discrepancies were addressed and resolved to ensure the codebook and themes accurately reflected the data. Furthermore, relationships within the codes and themes were explored, and a thematic map was created by HC, which was also discussed with the research team (Roberts et al., 2019). The results from the applied thematic analysis were shared with all participants for member checking to ensure the credibility of the data (Thomas, 2017; Motulsky, 2021). However, no comments were made and no changes were requested.

Table 1. Discussion guide for focus group discussion with 20 veterinarians, between July 2024 and April 2025[1]

Theme	Item	Question
Barriers to the uptake of education programs for veterinarians (30 min)	Read	For the purpose of this research, an education program can come in several forms, it can be virtual or in person, done as a group or can be self-guided such as a podcast or website and is designed to provide new information from a reputable source.
	Question 1	What kind of experience do you have with attending any type of education program yourself?
	Question 2	What is your overall impression of the education programs available for veterinarians working with dairy producers?
	Question 3	How does the content of these programs align with your day-to-day work?
	Question 4	What challenges, if any, have you faced when trying to access education programs for veterinarians?
	Question 5	What motivates you to participate in programs that allow you to continue your education?
		Follow up: Can you describe any reasons why you wouldn't take advantage of an education program that's offered to you?
	Question 6	How does the form that the education program is offered in impact your participation?
KTT barriers between veterinarians and dairy producers (30 min)	Read	According to dairy producers, veterinarians are one of the biggest influencers when making decisions regarding animal health as well as one of the main resources for offering or connecting producers with education programs.
	Question 7	What difficulties, if any, do you encounter when trying to transfer knowledge gained from education programs to dairy producers?
	Question 8	In your experience, how willing or interested are dairy producers in implementing veterinary advice based on new information from these programs?
	Question 9	What changes could be made to veterinary education programs to increase your participation and their impact on your work with dairy producers?
	Question 10	How, if at all, do communication barriers impact your experience associated with knowledge transfer for dairy producers?
	Question 11	How can education programs better foster collaboration between veterinarians and dairy producers to ensure more effective knowledge transfer?
	Read	Antimicrobial stewardship in animal agriculture is becoming a common topic as antimicrobial resistance continues to rise. As industry and academia identify strategies associated with responsible antimicrobial use, it is necessary that they are extended to dairy producers.
	Question 12	What is your experience with encouraging prudent antimicrobial use on farm, specifically with respect to calf disease?
	Question 13	Do you believe that treatment protocols for calf disease on farm contribute to prudent antimicrobial use?
General prompts	Prompt	Could you tell me more about that?
	Prompt	Could you walk me through that?
	Prompt	What do you think was going on there?
	Prompt	Does that often happen?
	Prompt	What do you normally do in that situation?
	Prompt	Would you mind explaining that again so that I can understand?
	Question 14	Is there anything anyone else would like to add?

[1]**Instructions—Read:** "There are no right or wrong answers to these questions. The purpose of this focus group is to gather opinions to understand the barriers associated with the uptake of education programs for veterinarians as well as barriers related to knowledge transfer and translation between veterinarians and dairy producers."

RESULTS

Participant Characteristics

Of the dairy producers, 13 were male and 7 were female, with the median (range) age of 38 (21 to 68) years. All participants lived in southwestern Ontario, Canada, and were either farm owners (90%) or managers (10%). The median (range) number of lactating cows in the herd was 94 (50 to 600), and the majority of producers operated in an automated milking system facility (55%), followed by parlor (30%) and tiestall barns (15%). Most producer participants had received postsecondary education (80%), while 20% of the participants' highest education level was high school.

For the veterinarians, 12 were male and 8 were female. The median (range) age of the participants was 49 (27 to 72) years, and the median (range) years practicing was 22.5 (2 to 51). Participants practiced in British Columbia (1), Alberta (3), Saskatchewan (1), Manitoba (1), Ontario (6), Quebec (3), New Brunswick (1), Nova Scotia (1), Prince Edward Island (2), and Newfoundland (1), Canada. Most of the participants were exclusively bovine (13), followed by mixed-practice (7) veterinarians.

Key Themes Identified

Four themes were identified using applied thematic analysis and are illustrated in Figure 1: (1) educational role of the veterinarian, (2) progress is a collaborative approach toward success, (3) veterinary strategies for knowledge transfer and translation, and (4) barriers to veterinary knowledge transfer and translation.

Theme 1: Educational Role of the Veterinarian. Both participating veterinarians and producers unanimously agreed that herd veterinarians play a significant role in

Table 2. Discussion guide for focus group discussion with 20 dairy producers, between July 2024 and April 2025[1]

Theme	Item	Question
Education program barriers and optimization (50 min)	Read	For the purpose of this research, an education program can come in several forms, for example, it can be virtual or in person, done as a group or can be self-guided and is designed to provide new information from a reputable source.
	Question 1	What kind of experience do you have with any type of education program?
	Question 2	How do you usually find out about new education programs?
	Question 3	What motivates you to participate in these education programs? Follow up: Can you describe any reasons why you wouldn't take advantage of an education program that's offered to you?
	Question 4	How does the content that's being offered affect your willingness to participate in the program? Education programs can be offered through several forms including in person, through webinars, podcasts, websites and phone applications. How does the form it is provided in impact your use of the program? Follow up: In your experience, how accessible are education programs for dairy producers in terms of scheduling, location, and technological requirements? Follow up: Are you more likely to use the program based on content or the form it is offered in?
	Question 5	What benefits have you experienced from the use of education programs?
	Read	Education programs are designed to provide resources that benefit dairy producers. There are likely to be programs offered that aren't necessarily suited to everyone's individual interests.
	Question 6	How likely are you to partake in an education program that aligns with your production system but does not align with your interests?
	Question 7	How do you perceive the usefulness of current education programs in improving farm practices and productivity?
	Question 8	In your experience, what would be the most effective way to enhance the uptake of education programs being offered to dairy producers?
	Question 9	What topics or content would you like to have accessible that you haven't seen offered yet?
General prompts	Prompt	Could you tell me more about that?
	Prompt	Could you walk me through that?
	Prompt	What do you think was going on there?
	Prompt	Does that often happen?
	Prompt	What do you normally do in that situation?
	Prompt	Would you mind explaining that again so that I can understand?
	Question 10	Is there anything anyone else would like to add?

[1]**Instructions—Have participants introduce themselves with first name and role on farm (while recording). Then read:** "The aim of our research is to understand the barriers related to the uptake of education programs for dairy producers and how stakeholders can improve their resources to encourage their use. Everything that is shared will be confidential, and you do not have to answer any questions that you do not want to. We expect this focus group to take about 45 minutes."

the education of dairy producers. Both groups perceived this educational role as a growing part of a veterinarian's responsibilities. Additionally, both sets of participants considered the importance of the producer-veterinarian relationship on client education. Three subthemes were identified: (1) veterinarians must be trusted, unbiased sources of knowledge; (2) filtering, interpretation, and validation of facts; and (3) building trusting relationships as a foundation for success.

Subtheme 1: Veterinarians Must Be Trusted, Unbiased Sources of Knowledge. Both groups of participants shared that they expect veterinarians to provide their clients with novel and applicable information: [P1_3] "your vet is one of the backbones of your business . . . they know your farm and the technical information to help you solve problems and improve." Additionally, it was highlighted that information coming from veterinarians is more likely to be unbiased: [V3_5] "As veterinarians, we aren't trying to sell them something like a pharmaceutical company rep would be. . . . I think our role is to give them that independent advice." It was suggested that the role of the veterinarian is shifting from clinician to advisor, focusing not only on animal health but also on providing important and novel information associated with overall farm management, further enhancing the value veterinarians bring to their clients:

> [V3_5] We are really moving toward the advisory role in large animal medicine, so I think that's where we can make the biggest difference for the farm.

> [P2_1] I'd say for me, my vet plays a pretty big role in education, whether it's because we're trying something new or they are helping make sure there's economic sense to changes that we're making.

As veterinary responsibilities shift, the value of continuing education (**CE**) was confirmed by participating veterinarians: [V4_2] "And so it's really important that you keep up, especially on the research side of things, because things change over time." Participating producers also agreed that CE for veterinarians was vital for the success of their clients' farms. They particularly believed

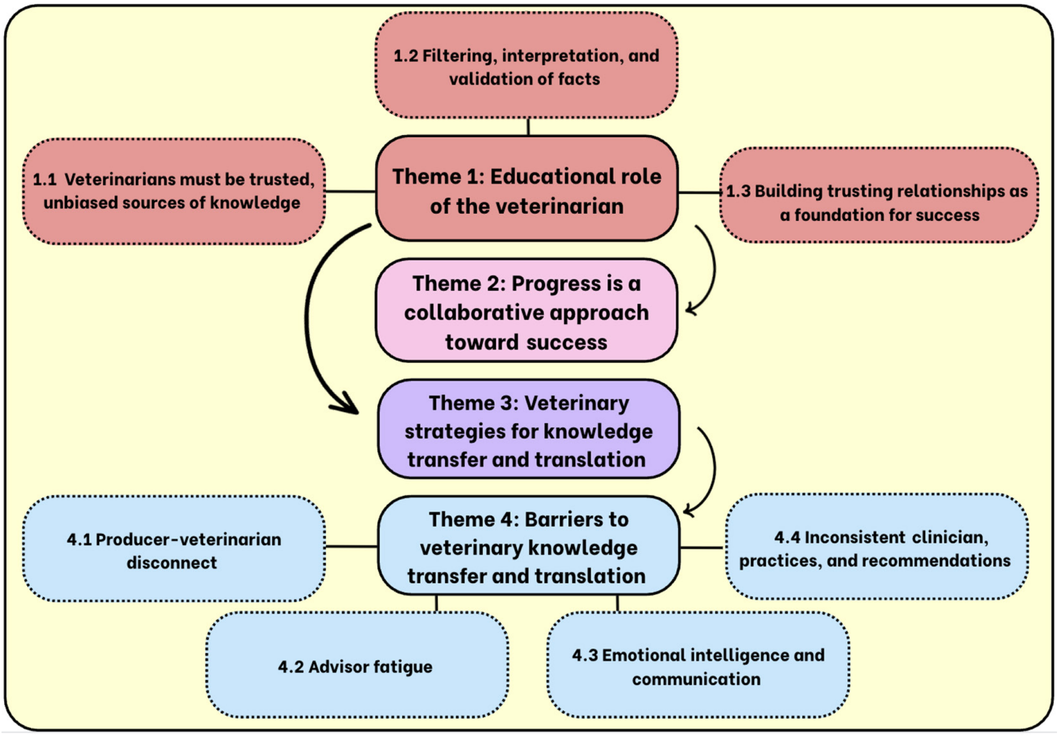

Figure 1. Thematic map outlining themes and subthemes identified through applied thematic analysis of 10 focus groups with 20 dairy producers and 20 veterinarians in Canada, between July 2024 and April 2025.

that veterinarians are putting more emphasis on their own CE, as highlighted by the following quote: [P5_1] "Veterinarians are just more focused on education outside of just the day-to-day stuff than they used to be."

Subtheme 2: Filtering, Interpretation, and Validation of Facts. Many learning opportunities are available to producers. However, several participating producers agreed that it is difficult to comprehend what content practically applies to individual farms, especially when all producers considered lack of time to be a primary barrier to education. Veterinarians are expected to inform producers how information may support their specific operation: [P2_5] "There's so much information we have access to, and there's not enough hours in the day to sift through it. So, I think relying on the professionals like veterinarians to sift through it first is huge. That has a lot of value." Moreover, veterinarians discussed that many of the information outlets producers have access to may not offer accurate information or valid messaging. As a result, the perceived role of the veterinarian is greater than knowledge transfer and translation and also includes validating the accuracy of the content: [V5_2] "There's more and more corporate influence on it [research]. . . . I worry about the amount of info that is available online . . . how accurate is it . . . you need someone that can tell producers the proper message from what is being presented."

Many participants in both groups agreed that peer learning between producers is valuable and effective: [V3_1] "Yeah, I think producers love to hear what other producers are doing and not just what we're recommending," and [P1_5] "sometimes your peers are the best people to talk to." However, participating veterinarians recognized that farm recommendations must be individualized to accommodate on-farm factors such as [V2_2] "infrastructure" and individual [V3_1] "producer goals." Learning from peers was compared with [V3_1] "playing that game telephone . . . they [producers] are getting more and more of their information from their peers, and often the source of that information is really sketchy." The sentiment of uncertainty was also highlighted by participating producers: [P3_5] "You start to realize there must be a reason why that other producer made a change to his calf protocols . . . was it to fix something, or was it just to improve? Sometimes you need to call in the professionals [veterinarians] to know if that's the way everybody should do it." This discussion highlighted the importance of communication between veterinarians and their clients to ensure new ideas are practical for their farms.

Subtheme 3: Building Trusting Relationships as a Foundation for Success. Producers discussed the value of working with their veterinarians: [P3_1]: "While we're doing herd health. . . . It's not just chatting; it's actually an hour of question time. We prepare ahead . . . to make sure we're maximizing that valuable time." Building and maintaining a strong relationship with their veterinarians was emphasized as a priority for farm success by both participating producers and veterinarians: [P1_4] "You need to build a good relationship with your vet," and [V4_1] "I think that comes down to client education, which I think is taken more seriously when you have a really good respectful relationship with your farmer." Participating veterinarians also highlighted the importance of gaining respect from their clients to encourage the adoption of their recommendations: [V4_1] "I think the respect from farmers, and the ability to communicate effectively with them, creates a good relationship, thereby earning their trust, and thereby being able to improve upon protocols and treatments on their farm, and then the cost benefit ratio is: I'm a good vet for you. I'm making your life better." Several qualities, such as [P3_1] "consistency," [P5_2] "frequent support," [P5_2] "transparency," and [P4_1] "empathy," were discussed as characteristics that establish respect for the veterinarian.

Theme 2: Progress Is a Collaborative Approach Toward Success. Both participating producers and veterinarians highlighted the need for improvement over time to enable success in their operations. Many participating producers felt that they had successfully progressed and aimed to continue to achieve considerable change on their own farms: [P2_1] "We're always trying to change to make the next step." However, it was also emphasized that change, or the perspective that a change is needed, is not often immediate: [P3_1] "I think it takes multiple times to catch on." This perspective was also shared by veterinarians: [V2_1] "I think that's an ongoing battle that may never end." Producers highlighted their reluctance to adopt new management practices in situations where they did not believe the change to be critical due to the perceived risk of failures such as economic losses or increased morbidity: [P3_5] "If you try and change something and it doesn't work, it's probably six months to a year to try and get back to where you were before." Participating veterinarians and producers agreed that a need must be felt to initiate a change: [P1_2] "Sometimes there's a topic they [the herd veterinarian] bring up, and it's like 'we're good' I'm not even interested," and [V4_3] "And if he doesn't perceive a need for change, I can talk for an hour, and the moment I'm driving off, he or she thinks, 'Oh, that was OK, but we're not really interested.' So there has to be that pain that makes them willing to change." Additionally, it was discussed that, along with producers, veterinarians must advance using CE to update their recommendations; however, this was perceived to be a considerable task: [V5_2] "It's a huge challenge educating both producers and vets. Not sure who's going to foot the bill for all that."

Accountability was reported as a crucial factor associated with implementing change and enabling success. Although veterinarians agreed that they are accountable for educating producers, it was emphasized that motivating change can be complicated: [V1_1] "As educators, that sometimes is a challenge when you're dealing with traditional producers that have always done something one way." Conversely, participating producers felt that their veterinarians should be held accountable for providing more information. It was reported that veterinarians and other advisors often incorrectly assume that producers already understand key information, which has limited the producers' ability to make informed decisions: [P3_4] "I'd say, if there's something that we really need to know, the vet needs to tell us, not just assume that we know what's going on." [P3_1] "Like I don't know why that [new information] is not pushed on us a little bit more so that we can make better decisions."

Theme 3: Veterinary Strategies for Knowledge Transfer and Translation. All participants highlighted that KTT between veterinarians and producers was vital for industry advancement. While KTT delivery methods were discussed, such as face-to-face communication and meetings, as well as written materials, it was widely recognized among both sets of participants that KTT is most effective when the information is individualized for each client: [V2_3] "So you kind of have to tackle people individually, or farms individually anyway," and [P3_3] "If the vet is using your own [performance] numbers, it's obviously going to resonate more." Offering similar content in multiple delivery methods was also emphasized when discussing strategies to enhance uptake: [V3_4] "We need to diversify our sources of information for our clients. There's us directly but also pamphlets and written articles."

Participating veterinarians discussed applying several different KTT strategies to share information with producers. Both veterinarians and producers perceived hands-on training and small group meetings to be more effective for achieving progress by targeting interested individuals; however, it was noted that larger annual meetings should not be overlooked, as they are more inclusive and help spread broader awareness for more general topics. Additionally, it was highlighted by veterinarians that collaborating with other on-farm advisors, such as nutritionists, could amplify the effectiveness of knowledge transfer by illustrating that the topic is of high importance and that advisors' goals are aligned. Similarly, it was reported that hosting meetings with external industry actors was useful when a message was expected to be perceived as more valuable or more credible coming from a third party, for example, hoof trimmers and nutritionists. Several participating veterinarians also considered these collaborative meetings as learning opportunities for themselves. Furthermore, many participating veterinarians shared that social media is becoming a successful KTT tool to reach a wider audience; sharing brief educational content, such as calving pictures or new products, gained enthusiasm from clients. It was perceived that most social media users were younger producers. Additionally, inviting clients to CE developed for veterinarians was suggested as an effective KTT strategy; however, it was noted that it is difficult to offer this to many producers at one time. The 2 wk following CE attendance were considered to be an optimal time to share any newly learned information with producers, otherwise, “you sort of lose track of it” [V3_2].

It was also emphasized by participating veterinarians that effective KTT not only depends on the content but also on how it is delivered. It is vital to offer information in a manner that stimulates interest and adoption by each producer: [V1_3] “I really like to consider what is most important for certain farmers, what the angle of motivation is for them.” Several veterinarians also acknowledged that designing and delivering KKT can be challenging, noting that there are likely more effective KTT tools and strategies available that they don’t know about: [V3_1] “I think there’s lots of tools out there, but I’d say that I probably don’t have the training to use them to their full effect.”

Theme 4: Barriers to Veterinary Knowledge Transfer and Translation. While numerous KTT strategies were reported and used, several barriers were highlighted and should be considered to ensure information is extended effectively for uptake and implementation. In both producer and veterinarian focus groups, intrinsic and external barriers were discussed: (1) producer-veterinarian disconnect, (2) advisor fatigue, (3) emotional intelligence and communication, and (4) inconsistent clinicians, practices, and recommendations.

Subtheme 1: Producer-Veterinarian Disconnect. Many participants described a disconnect between producers and veterinarians. Veterinarians discussed that producers have many outlets to gain new information and insights, which are often shared with them. One veterinarian shared: [V2_1] “Often our producers have information at the same time or sooner than we do, and they’re the ones that are bringing it forward. And it does kind of make us look dumb in the moment, which is tricky.” Conversely, certain producers shared that they feel the support from their herd veterinarian is either too infrequent due to both the veterinarians’ and producers’ busy schedules or [P3_5] “too expensive.” As a result, participants reported it was difficult to rely on their veterinarians for additional support beyond routine visits; both of these factors were said to diminish the relationship:

> [P3_5] “Now you can have heat detection systems that can pretty much preg check. I went from herd health every two weeks to every six weeks. . . . The vet has less time at your farm, so he knows a little less about what’s going on, and then you lose that relationship . . . because he loses touch with what’s going on on your farm.”

It was highlighted that this disconnect affects the trust participating producers have in their veterinarians and, as a result, reduces the influence their veterinarian has on their farm: [P1_1] “It’s important to use your vet, but sometimes you have to use your own intuition, I know my farm best.” Furthermore, this strain on their relationship reduces veterinarians’ KTT abilities: [P2_5] “So yeah, then you start to mistrust the people that are supposed to be educated and are supposed to be educating you.” However, not all veterinarians recognize this barrier: [V3_2] “But we have to really keep that role [trusted advisor] because as we’re making fewer and fewer calls on these farms and mostly moving into an information and consulting role, and we need to really take that role to heart and become the main information source for them.”

Subtheme 2: Advisor Fatigue. Several participating veterinarians shared that KTT can be discouraging if they do not see any changes being adopted. This sentiment was perceived as a reflection that their KTT efforts were ineffective. One participant shared: [V3_4] “Then you leave thinking the client understood, we created a nice treatment protocol, and then it ends up in a binder with a bunch of fly droppings on it.” Some veterinarians considered this to give them a sense of [V3_2] “wasted time” and wanting to [V3_2] “give up on the client.” For other participating veterinarians, advisor fatigue transformed how they measured success: [V2_3] “It makes me feel a little bit better that I have tried to help them out. If they don’t want to listen, then that’s up to them, but at least it makes me feel like I did something.”

Subtheme 3: Emotional Intelligence and Communication. Discussions on enacting change raised the importance of emotional intelligence for veterinarians. Both groups highlighted that soft skills, including self-awareness, social awareness, and empathy, are salient characteristics for developing relationships and effective communication with their clients: [V3_1] “soft skills and communication are hugely important in this job,” and [P2_5] “That’s where I close down. Sometimes when you’re having an issue, they [the veterinarian] point at the 100 things that you’re probably doing wrong. It’s like, okay, are there things that can be improved? Yes, but that’s not at the root of my problem. If they need me to be receptive . . . they need to get to the root of the problem with me." However, it was also acknowledged by veterinarians that these traits are difficult to learn and not often taught: [V5_2] "I don't know why we go through vet school, and we don't get a course on how to relate with producers and how to educate producers." As a result, it was emphasized that emotional intelligence differs among individual veterinarians: [V3_3] "Some vets don't have a clue what the farmer wants. They're just still thinking about veterinary science, not even veterinary medicine. And that's my fear . . . sometimes maybe I'm missing the point, right? I need someone to help redirect me as to what is important."

Many participating veterinarians were concerned with their emotional intelligence and that of their peers, relating to their ability to effectively perceive and respond to emotion. Among younger, early-career veterinarians, these concerns were often connected to their struggle with imposter syndrome. Participating veterinarians misreported feeling underqualified and poorly equipped to support producers, specifically in demonstrating confidence: [V3_1] "There's a lot of imposter syndrome coming onto a farm, especially if you don't feel like you have a lot of real-world experience." It was suggested that participating in CE, specifically "train the trainer" initiatives, is a valuable strategy to enhance certainty: [V3_1] "It can be useful to already have tools in place, and to have training for trainers, so people [veterinarians] feel more comfortable communicating an important message."

Subtheme 4: Inconsistent Clinicians, Practices, and Recommendations. Both participating producers and veterinarians shared perspectives on variability in practice. Producers felt that there was a noticeable difference among veterinary clinics and individual veterinarians related to what KTT and expertise are offered to their clients: [P3_1] "I think you have to remember that all vets are not . . . equal. That can have a big impact on what [information] they bring to our farm." Producers agreed that the veterinarians' age, experience, use of technology, and education all affected their KTT experiences: [P2_1] "The vet that came in that day wasn't our usual vet, and she spoke to me as if I knew this information when in reality, my vet had never told me anything like it." Furthermore, certain participating veterinarians illustrated the lack of consistency among recommendations: [V5_1] "I think it's very important to develop SOPs in multi vet practices, so no matter who goes [to the farm] we're all on the same page so we're not hearing, 'Well how come you're saying this?' and 'Doctor So-and-So does the opposite?' . . . We need to try very hard to practice evidence-based medicine. And we do that by trying to keep up and being on the cusp, so to speak." Both producers and veterinarians agreed that variability can be detrimental to credibility and trust: [P3_5] "If there's a real issue here that needs to be dealt with and they send someone different, how do I know if they did the right thing, especially when I know my vet would have fixed it differently? You start to question who would have been more right."

DISCUSSION

Through this qualitative study, we explored participating dairy producers' and veterinarians' perceptions of the educational role their herd veterinarian holds and the challenges associated with KTT. Our main finding was that veterinarians are expected to be trusted educators who should enable producers' success. While several educational strategies were considered, many barriers to KTT exist from both the veterinarians' and producers' perspectives. These interrelated findings were illustrated in 4 main themes: (1) the educational role of the veterinarian, (2) progress is a collaborative approach toward success, (3) veterinary strategies for knowledge transfer and translation, and (4) veterinary barriers to knowledge transfer and translation.

Veterinarians as Educators

Participating dairy producers and veterinarians believe that education and advice are growing components of a veterinarian's responsibility to their clients. This notion is consistent with previous findings where producers considered veterinarians to be a valuable source of management information and guidance (Lam et al., 2011; Ritter et al., 2020; Cobo-Angel et al., 2021). While participants agreed that veterinarians should be educators, several factors affect a producer's perception of the information. Most participants noted considerable variation in the information provided by different veterinarians and clinics. Similar findings have been observed in past research, where thresholds for disease recording and treatment differed among bovine veterinarians (Espetvedt et al., 2013; Cobo-Angel et al., 2023). While predictable veterinarian behavior has been identified as a vital pillar for a strong relationship (Bard et al., 2019), participating producers expressed that inconsistent recommendations undermined their trust in veterinary advisors. Developing and implementing strategies to standardize recommendations could help strengthen this trust. For example, accredited CE has been identified as an effective strategy to make recommendations more consistent and improve practice (Kareskoski, 2023). In addition to consistency, frequency of support was considered to be a necessary factor to foster a strong connection between the producer and veterinarian. Winder et al. (2015) found that increasing the frequency of routine herd visits was associated with improved adoption of pain management strategies when dehorning and disbudding. Further, frequent interaction and support are characteristics of well-established veterinarian-producer relationships (Gröndal et al., 2023). While several participating producers were concerned with infrequent visits, this was not as commonly discussed among participating veterinarians. One strategy may be to explicitly repurpose or at least refocus routine herd health appointments to formulate management practices and plans together (Ritter et al., 2021).

Communication and Soft Skills Are Needed to Support Producer Success

Both participating producers and veterinarians reported that progress at the farm level elicits industry success; however, it seemed that participating producers were more confident in their progress than participating veterinarians. Progress may be conceptualized differently by each group, suggesting a misalignment of goals where producers may be driven by farm-specific advancements, while veterinarians may view progress as industry improvement (Ciaravino et al., 2023). Inconsistent perspectives have also been identified in previous research. Although producers and veterinarians share similar concerns associated with disease and pain management, they disagree on beliefs about disease prevalence and pain (Sumner et al., 2018a).

These differences may be partially attributed to the reported criticism of assumption-based communication, where veterinarians or other advisors make recommendations while assuming client knowledge and priorities without confirming alignment or understanding (Carter et al., 2026). This phenomenon leads to miscommunication between veterinarians and clients, which is likely to result in differing perspectives and an inability to align and discuss due to the risk of discouraging producers from seeking clarification and engaging further in the topic (Janke et al., 2021; Carter et al., 2026). Veterinary communication tools, such as asking open-ended questions to encourage in-depth responses and promoting clarifying questions from producers to ensure alignment of plans and goals, can limit the risk of misunderstanding (Dysart et al., 2011; Janke et al., 2021). Efforts should also be made to improve communication skills with peers through training in authentic settings, for example, on routine farm calls (Vleuten et al., 2019). Communication is a growing focus in Canadian and other veterinary curricula (Mossop et al., 2015), but criticisms were still raised by early-career veterinary participants calling for even greater emphasis on this topic while completing their degree. This highlights that efforts are being made to support veterinarians, but they may not yet be sufficient.

Veterinarians highlighted several KTT strategies that they had implemented, including group meetings and written materials; however, both groups agreed that individualized content is more valuable for producers. Individualizing content for each client may not be achievable when the time allocated to KTT is limited. Although group meetings are more efficient for veterinarians and highly valued by producers, it is often difficult for producers to find a personalized take-home message from the content (Carter et al., 2026). Efforts should focus on identifying time-efficient yet individualized methods to extend information. For example, if the content allows, benchmarking can be incorporated into group meetings, allowing producers to anonymously compare themselves to their peers. Additionally, they can use this information to encourage change or prompt questions to direct to their veterinarian (Sumner et al., 2018b). Additionally, small group learning facilitated by farm advisors such as veterinarians has been shown to help producers individualize topics through peer discussion and instructor support (Carter et al., 2026). Furthermore, follow up is vital after initial knowledge extension to support accountability and motivation and to assess on-farm feasibility, offering the opportunity for collaborative adjustments if needed (Ritter et al., 2019; Svensson et al., 2019).

Social media was reported as an outlet to interact and stimulate discussion with producers. Its flexibility may be advantageous for accommodating varied schedules. Veterinarians are often busiest midday when producers tend to have more flexibility (Carter et al., 2026). To optimize outreach, veterinarians could create and schedule social media content to be posted when producers are more active online. This approach is particularly effective for engaging younger producers, who are more likely to use digital platforms. However, it is important to recognize that social media has been deemed a less important source of information compared with other KTT methods (Roche et al., 2020). Therefore, it should be used in addition to other strategies, such as in-person presentations and written materials (Carter et al., 2026).

Effective communication is fundamental to maintaining strong client relationships, fostering trust, and promoting adoption of recommendations. It has been shown to improve clinical outcomes, client satisfaction, and compliance (Artemiou et al., 2014). As a result, many veterinarians consider communication skills to be as important, or even more important. than clinical expertise (McDermott et al., 2015; Morabito et al., 2025). Yet, communication emerged as a recurring theme that was connected to several barriers to KTT, including producer-veterinarian disconnect, advisor fatigue, limited emotional intelligence, and inconsistent messaging. There are strong linkages between a lack of soft skills and communication and imposter syndrome, a feeling reported by participating veterinarians as well as in previous research in veterinary and human medicine (Kogan et al., 2020). Several veterinarians in this study emphasized the need for enhanced communication training within veterinary education programs. Additionally, advisor fatigue was often discussed by participating veterinarians, and participating producers shared that expectations of veterinary engagement were not continuously met, which may have been connected to advisor fatigue and ultimately, the reduced opportunity for positive change. Continuing education focused on communication has been recommended to strengthen veterinarians' ability to effectively engage clients (Pun, 2020). Such training should include both verbal and nonverbal communication, as nonverbal cues can substantially influence meaning and, when misinterpreted, lead to misunderstanding between veterinarians and clients (Carmichael and Mizrahi, 2023). Enhanced communication education will support veterinarians in their professional careers; however, it must be emphasized that CE associated with these skills will also support veterinarians' mental health (McGregor et al., 2008).

Study Limitations and Considerations

We acknowledge some limitations to this study. Recruitment for veterinarian focus groups was conducted nationally to achieve an adequate number of participants, and as a result, focus groups were conducted virtually. Although virtual focus groups limit nonverbal observations (Hassell and Cotton, 2017), all participants reported confidence in the technical setup of video conferencing and used these functions, thus reducing this limitation (van der Kleij et al., 2009). Producer focus groups were conducted with dairy producers in Ontario. Southern Ontario is one of the most concentrated dairy production areas in Canada (Agriculture and Agri-Food Canada, 2024); therefore, it is anticipated that KTT barriers observed here are likely also to be observed among producers in other regions of Canada, with more isolated regions possibly experiencing similar barriers to a greater extent. Furthermore, the mean age of dairy producers in Ontario is higher than that of the study participants (Agriculture Canada, 2021); however, we are confident that the wide age range of participants captured perspectives to represent each demographic. Veterinarians who volunteered to participate may have been particularly well-positioned to provide insight due to their interest in communication and knowledge transfer compared with the generalized veterinary population. Although this may have influenced the emphasis of knowledge mobilization barriers, the reported concerns have been well documented in previous literature and likewise were consistent among participating veterinarians of varying ages, years of practice, and clinic types. This suggests that while the participants may have been outliers in their interest in the subject, their perspectives were likely shared within the veterinary system. Additionally, less interested veterinarians may have had fewer insights, experiences, or motivations associated with KTT, thereby limiting the depth of discussion that was achieved during the focus groups within this study. Conducting focus groups with veterinarians and producers together may have gained different insights; however, aside from logistical considerations, we chose to separate groups to encourage a comfortable environment for candid and open discussion of barriers and weaknesses to KTT from individual perspectives without fear of judgment or tension between the groups (Noyes, 2022). Furthermore, there is a risk that convenience and snowball sampling reached more interested, similar-minded, or extroverted participants, but we believe that sending invitations through herd veterinarians and peers was our best effort to reach less engaged participants.

The results from this study cannot be generalized to a broader audience, and therefore, additional research in this area is encouraged to continue to explore and facilitate knowledge mobilization between veterinarians and dairy producers. Nonetheless, we are confident that data saturation was adequately achieved for meaningful coding of the data under the identified themes, as no new themes were identified in the last 2 producer focus groups or in the final veterinarian focus group (Saunders et al., 2018). Additionally, pairing the dense sample specificity, strong dialog within focus groups, and the clarity of the research question collectively support our confidence in adequate information power (Malterud et al., 2016). The combination of data saturation and information power reinforces that the findings of this study sufficiently address and answer our research question. Moreover, no participant disagreed with the representation of their perspectives in the final results.

CONCLUSIONS

Participating producers and veterinarians both expressed the importance of the herd veterinarian's role in education, particularly validating information offered through other sources. Adoption of KTT by participating producers was dependent on trust and the strength of the relationship between them and their veterinarian, developed through consistent and clear communication. However, communication skills and time constraints were reported to be barriers to effective KTT. Education is a catalyst for change and progress. Therefore, it is vital that KTT is delivered using effective strategies to mitigate these barriers. For example, offering benchmarking content in group meetings to help producers individually reflect, and including communication and KTT content in veterinarian CE to share with producers. The impact of a scientific study is not only contingent on the research itself but also on the effectiveness of its associated KTT. The findings of this study emphasize that strategically designed and delivered KTT practices are essential to ensure research is both accessible and practical for dairy producers.

NOTES

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Nonstandard abbreviations used: CE = continuing education; KTT = knowledge transfer and translation; P = producers; V = veterinarians.

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